

Culinary Corridors: Mapping Where Food Similarity Breaks Distance

Run 4 Winner-Aligned Fisher Report

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Abstract

Culinary Corridors uses GIS to ask when cuisine similarity follows geographic distance and when it breaks distance-based expectation. Cuisines are represented as filtered ingredient profiles, pairwise similarity is calculated, and a distance baseline is fit using approximate geographic anchors. The residuals become candidate culinary corridors. The final package uses a global discovery screen, an East/Southeast Asia primary case, and a residual bridge-index analysis to show that cuisine similarity has spatial structure that cannot be found through ingredient clustering alone. The project makes cautious, non-causal claims and treats migration, trade, colonial, and maritime explanations as hypotheses for future testing.

1 Fisher-aligned question

The winner-aligned research question is: **when does cuisine similarity follow geographic distance, and when does it exceed distance-based expectation?** This reframing moves the project away from a general food-data study and toward a spatial inference project. A non-spatial analysis could cluster similar cuisines, but it could not identify which similarities are unexpected relative to distance, where those residuals lie, or which places become bridge nodes in the residual geography.

2 Method

The workflow has four core steps: ingredient-profile construction, pairwise similarity calculation, geographic distance baseline, and residual mapping. Positive residuals are candidate culinary corridors. A bridge-index upgrade aggregates residual links into mapped place-level roles.

3 Focused case

The global model is a discovery screen. The strongest interpretation comes from East/Southeast Asia, where retained cuisines form a coherent regional and cross-subregional geography. The focused case avoids the overreach of treating the global map as a causal model.

4 Geospatial-only insight

The Run 4 bridge-index upgrade converts pairwise residual links into mapped place-level roles. This is the clearest Fisher-style insight because it depends on geography, distance, and residual structure; it cannot be reproduced from ingredient vectors alone.

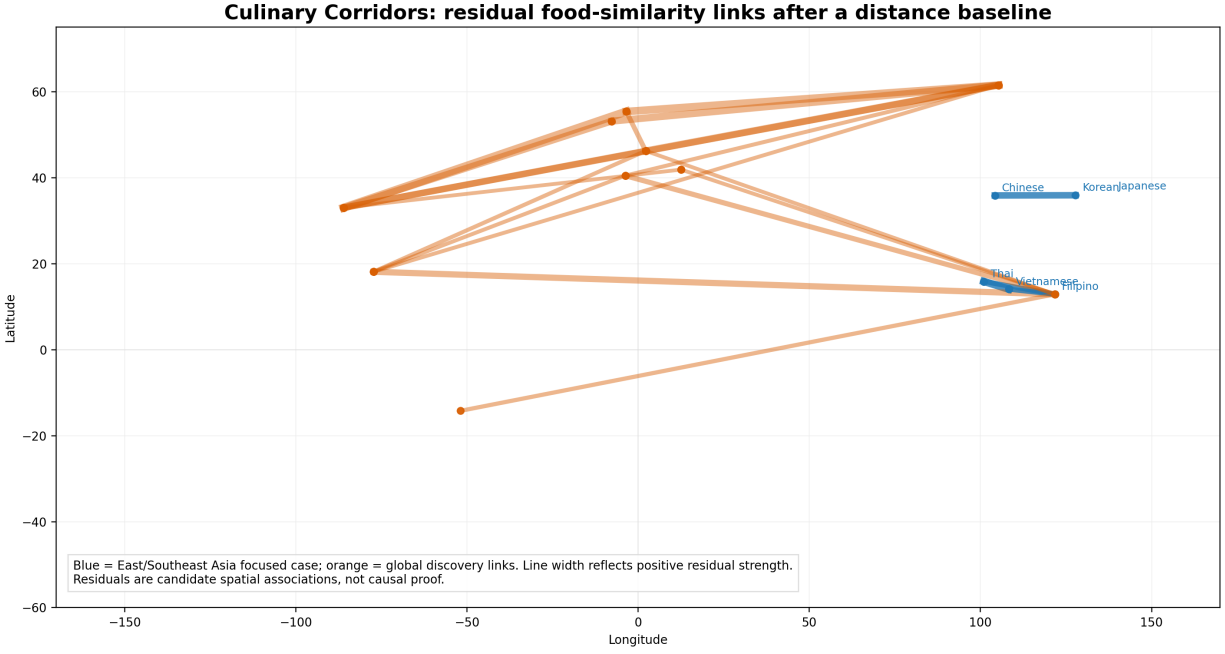


Figure 1: Hero spatial argument. Positive residual culinary corridors after a distance baseline. Blue links highlight the East/Southeast Asia focused case; orange links show global discovery links. Line width reflects residual strength.

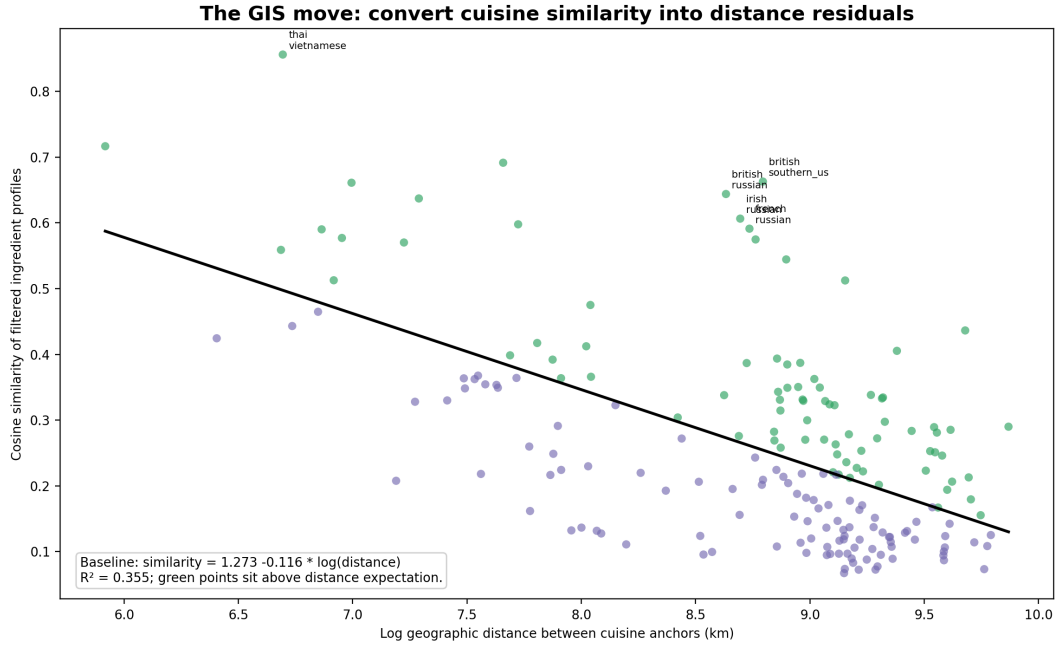


Figure 2: Method figure. Cuisine similarity is compared to log geographic distance. Points above the fitted baseline have positive residuals and become candidate corridors.

5 Limitations and claim discipline

The recipe corpus is platform-mediated and should not be treated as a representative census of world cuisine. Cuisine-to-place mapping is approximate. Residual corridors are model outputs, not causal

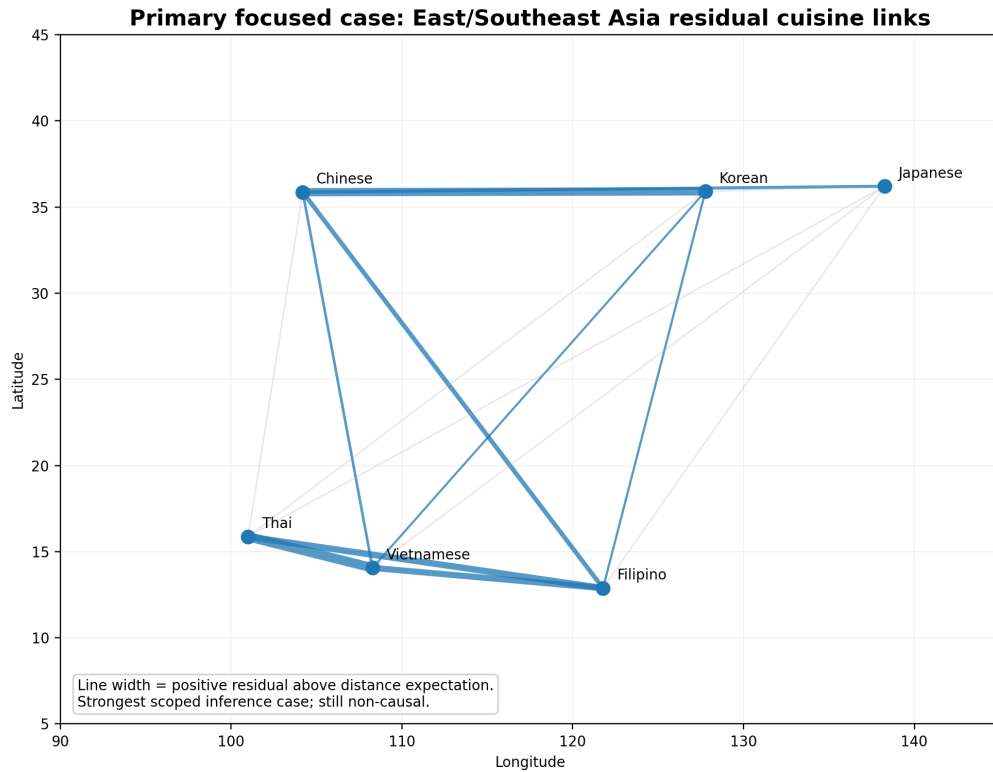


Figure 3: Primary focused case. East/Southeast Asia residual cuisine links after filtering generic ingredients. Thicker links indicate stronger positive residuals.

historical facts. The boundary/permeability diagnostic helps preserve this discipline by showing where residual similarity differs across spatial groupings without converting those groupings into causal explanations.

6 Before/after Fisher alignment

The Run 4 revision improves the project by sharpening the thesis, strengthening the hero figure, foregrounding the residual GIS operation, adding a bridge-index spatial role metric, and rewriting the submission around map-led evidence. The internal playbook score improves from 80.8 to 91.9 under the packet's weighted criteria, pending manual StoryMap/poster QA and data-license confirmation.

7 Submission recommendation

The recommended route remains a StoryMap or web-map-heavy submission, with a compliant poster text draft available as a backup. The final submission should lead with the hero residual map, explain the residual method, focus on East/Southeast Asia, use bridge roles as the geospatial-only insight, and close with explicit limitations.

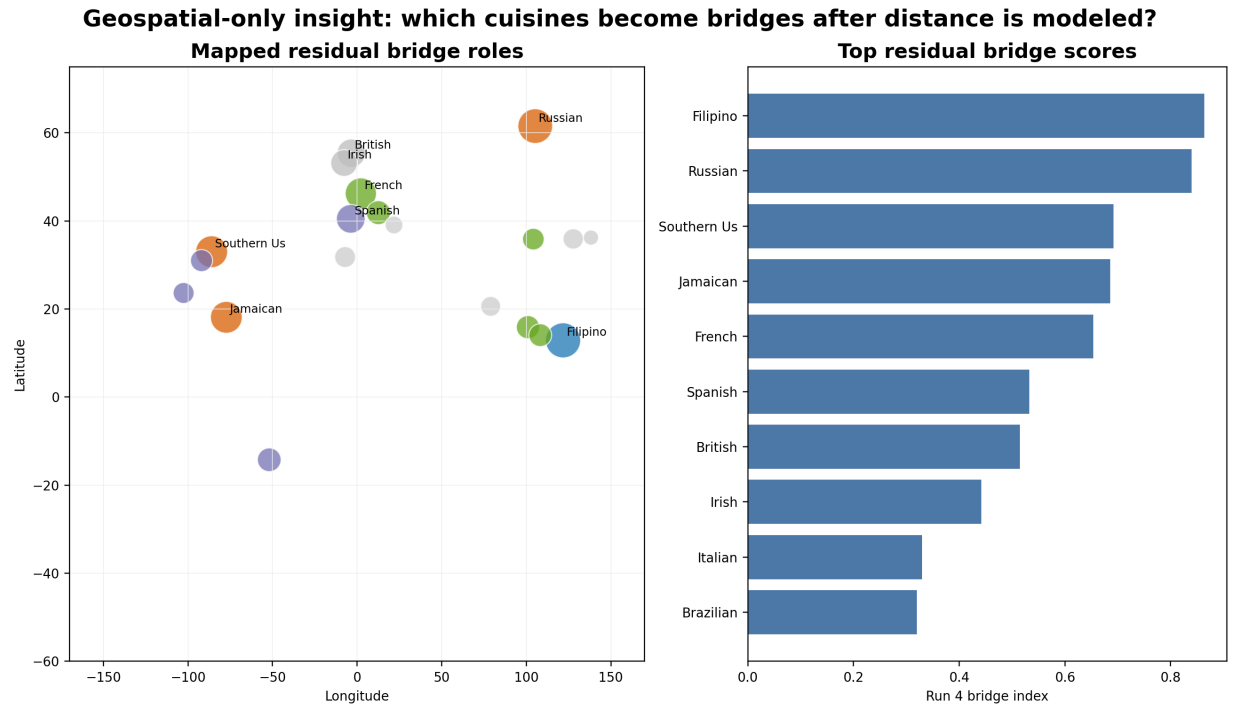


Figure 4: Residual bridge roles. The bridge index aggregates positive residual degree, top-residual participation, mean residual strength, long-distance residual score, and mean residual across all dyads.

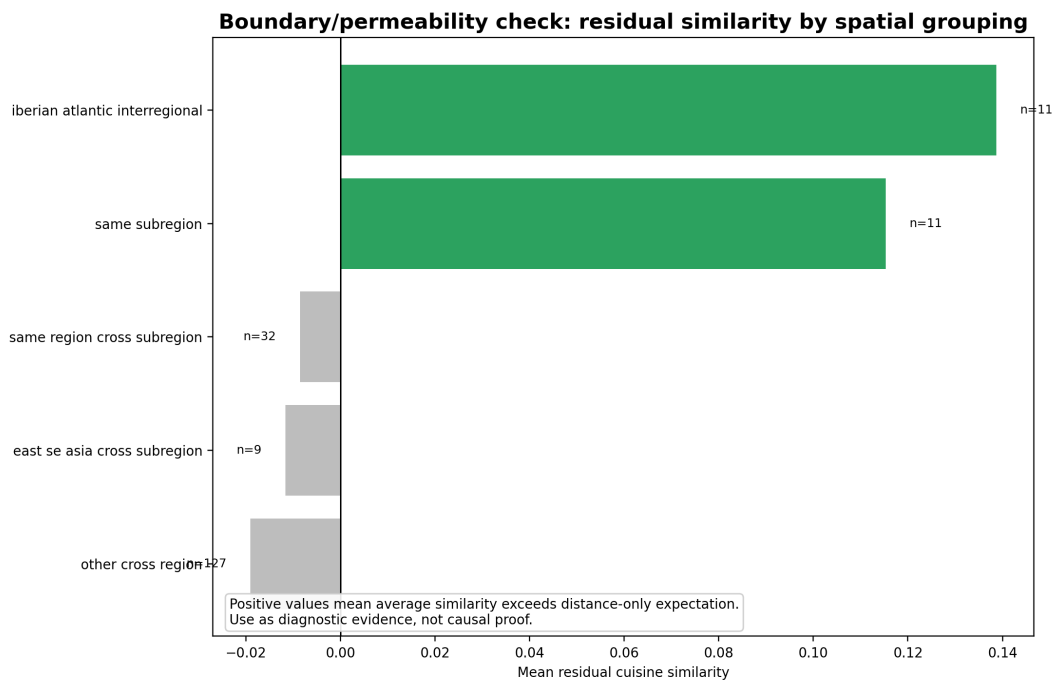


Figure 5: Boundary/permeability diagnostic. Mean residual cuisine similarity by spatial grouping. Positive values indicate similarity above the distance-only expectation.